

AI based research paper summarization system - LAB Exercise – 3

1. Write SQL statements to demonstrate the following JOIN operations on the tables created for your application domain. INNER, LEFT, RIGHT, and FULL OUTER JOINS.

Specifications: a. The operations should be performed on tables with logical sense and mention valid justification. b. What is the difference between each JOIN?

PART 1 - JOIN OPERATIONS:

- INNER JOIN

```
101 • SELECT
102     rp.PaperID,
103     rp.Title,
104     u.FullName,
105     u.Role
106 FROM ResearchPaper rp
107 INNER JOIN User u
108 ON rp.UserID = u.UserID;
```

Result Grid

	PaperID	Title	FullName	Role
▶	101	AI in Healthcare	Alice Johnson	Student
	102	Deep Learning Basics	Bob Smith	Researcher
	103	Natural Language Processing	Charlie Brown	Student
	104	Computer Vision	David Wilson	Researcher
	105	Machine Learning Algorithms	Emma Thomas	Student
	106	Cloud Computing	Frank Miller	Researcher
	107	Big Data Analytics	Grace Lee	Student

- Displays research papers along with the user who uploaded them.
- Only matching records from both tables are returned

- **LEFT JOIN**

```
109
110 • SELECT
111     u.UserID,
112     u.FullName,
113     rp.Title
114 FROM User u
115 LEFT JOIN ResearchPaper rp
116 ON u.UserID = rp.UserID;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	UserID	FullName	Title
▶	1	Alice Johnson	AI in Healthcare
	2	Bob Smith	Deep Learning Basics
	3	Charlie Brown	Natural Language Processing
	4	David Wilson	Computer Vision
	5	Emma Thomas	Machine Learning Algorithms
	6	Frank Miller	Cloud Computing
	7	Grace Lee	Big Data Analytics

- Displays all users.
- If a user has not uploaded any paper, the paper title will be NULL.

- RIGHT JOIN

```
117
118 • SELECT
119     u.FullName,
120     rp.Title
121 FROM User u
122 RIGHT JOIN ResearchPaper rp
123 ON u.UserID = rp.UserID;
```

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	FullName	Title			
▶	Alice Johnson	AI in Healthcare			
	Bob Smith	Deep Learning Basics			
	Charlie Brown	Natural Language Processing			
	David Wilson	Computer Vision			
	Emma Thomas	Machine Learning Algorithms			
	Frank Miller	Cloud Computing			
	Grace Lee	Big Data Analytics			

- Displays all research papers.
- If a paper has no matching user (unlikely because of the foreign key), the user details will be NULL.

- FULL OUTER JOIN

```

125 • SELECT
126     rp.PaperID,
127     rp.Title,
128     s.SummaryID,
129     s.SummaryText
130 FROM ResearchPaper rp
131 LEFT JOIN Summary s
132 ON rp.PaperID = s.PaperID
133 UNION
134 SELECT
135     rp.PaperID,
136     rp.Title,
137     s.SummaryID,
138     s.SummaryText
139 FROM ResearchPaper rp
140 RIGHT JOIN Summary s
141 ON rp.PaperID = s.PaperID;

```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	PaperID	Title	SummaryID	SummaryText
▶	101	AI in Healthcare	201	Healthcare AI summary
	102	Deep Learning Basics	202	Deep Learning summary
	103	Natural Language Processing	203	NLP summary
	104	Computer Vision	204	Computer Vision summary
	105	Machine Learning Algorithms	205	Machine Learning summary
	106	Cloud Computing	206	Cloud Computing summary
	107	Big Data Analytics	207	Big Data summary

- This query displays **all research papers** and **all summaries**.

- Difference Between JOINS:

JOIN	Returns
INNER JOIN	Returns only the matching records from both tables.
LEFT JOIN	Returns all records from the left table and matching records from the right table.
RIGHT JOIN	Returns all records from the right table and matching records from the left table.
FULL OUTER JOIN	Returns all records from both tables.

2. Demonstrate the usage of subqueries to implement the operations to be performed on the tables created for your application domain. Specifications: a. At least 5 subqueries should be demonstrated. b. All tables should be used in at least 1 Compound SQL Query to perform a specified operation. c. The Operations should be performed on tables with logical sense and mention valid justification.

PART 2 – SUBQUERIES

1. Papers uploaded by Researchers

```
143 • SELECT PaperID, Title
144 FROM ResearchPaper
145 WHERE UserID IN
146 (
147 SELECT UserID
148 FROM User
149 WHERE Role = 'Researcher'
150 );
```

PaperID	Title
102	Deep Learning Basics
104	Computer Vision
106	Cloud Computing
NULL	NULL

- Displays all research papers uploaded by users whose role is **Researcher**.

2. AI Models with Accuracy above Average

```
152 • SELECT ModelID, ModelName, Accuracy
153 FROM AIModel
154 WHERE Accuracy >
155 (
156 SELECT AVG(Accuracy)
157 FROM AIModel
158 );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	ModelID	ModelName	Accuracy
▶	2	GPT-4	98.10
	6	LLaMA	97.00
	7	Gemini	98.40
•	NULL	NULL	NULL

- Displays AI models whose accuracy is greater than the average accuracy of all AI models.

3. Users Who Have Given Feedback

```
159
160 • SELECT UserID, FullName
161 FROM User u
162 WHERE EXISTS
163 (
164 SELECT *
165 FROM Feedback f
166 WHERE u.UserID = f.UserID
167 );
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	UserID	FullName
▶	1	Alice Johnson
	2	Bob Smith
	3	Charlie Brown
	4	David Wilson
	5	Emma Thomas
	6	Frank Miller
	7	Grace Lee
•	NULL	NULL

- Displays users who have submitted feedback.

4. Papers Having Long Summaries

```
169 • SELECT Title
170 FROM ResearchPaper
171 WHERE PaperID IN
172 (
173 SELECT PaperID
174 FROM Summary
175 WHERE SummaryLength > 130
176 );
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Title			
▶	Deep Learning Basics			
	Computer Vision			
	Cloud Computing			
	Big Data Analytics			

- Displays research papers whose summary length is greater than 130 words.

5. Users Who Have Not Uploaded Any Paper

```
178 • SELECT UserID, FullName
179 FROM User u
180 WHERE NOT EXISTS
181 (
182 SELECT *
183 FROM ResearchPaper rp
184 WHERE rp.UserID = u.UserID
185 );
```

Result Grid		Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	UserID	FullName			
*	NULL	NULL			

- Displays users who have not uploaded any research paper.

➤ Compound SQL Query

```
187 • SELECT
188     u.UserID,
189     u.FullName,
190     u.Role,
191     r.Bio,
192     rp.Title,
193     a.ModelName,
194     su.SummaryText,
195     sh.Notes,
196     f.Rating
197 FROM User u
198 LEFT JOIN Student st
199 ON u.UserID = st.StudentID
200 LEFT JOIN Researcher r
201 ON u.UserID = r.UserID
202 LEFT JOIN ResearchPaper rp
203 ON u.UserID = rp.UserID
204 LEFT JOIN Summary su
205 ON rp.PaperID = su.PaperID
206 LEFT JOIN AIModel a
207 ON su.ModelID = a.ModelID
208 LEFT JOIN SummaryHistory sh
209 ON su.SummaryID = sh.SummaryID
210 LEFT JOIN Feedback f
211 ON rp.PaperID = f.PaperID
212 WHERE u.UserID IN
213 (
214     SELECT UserID
215     FROM User
216     WHERE Role = 'Researcher'
217 );
```

Result Grid								
Filter Rows:								
Export: Wrap Cell Content:								
	UserID	FullName	Role	Bio	Title	ModelName	SummaryText	Notes
2	Bob Smith	Researcher	AI Researcher specializing in Machine Learning	Deep Learning Basics	GPT-4	Deep Learning summary	Useful	4
4	David Wilson	Researcher	Expert in Natural Language Processing	Computer Vision	RoBERTa	Computer Vision summary	Needs Improvement	3
6	Frank Miller	Researcher	Researcher in Deep Learning and Neural Networks	Cloud Computing	LLaMA	Cloud Computing summary	Very Good	5

- Uses **all eight tables** (User, Student, Researcher, ResearchPaper, AIModel, Summary, SummaryHistory, Feedback).
- Uses a **subquery (IN)** to retrieve only users whose role is **Researcher**.